



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
21-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-1

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION- I(i) : (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

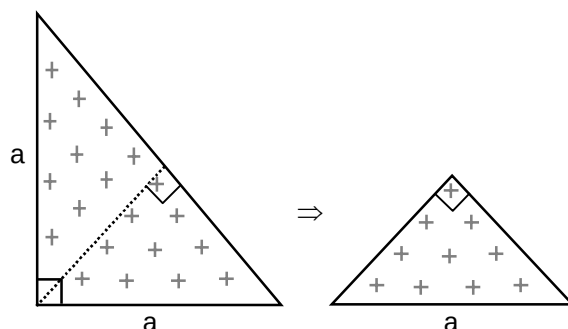
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

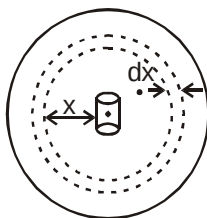
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1. The radiations emitted by an excited hydrogen atom are observed at an angle of 45° with respect to its direction of motion. Calculate its velocity if the first line of lyman series shows a shift of $\Delta\lambda = 0.2 \text{ nm}$ [Rydberg constant = $1.097 \times 10^7 \text{ m}^{-1}$]
- (A) $6 \times 10^5 \text{ m/s}$ (B) $5 \times 10^5 \text{ m/s}$ (C) $7 \times 10^5 \text{ m/s}$ (D) $8 \times 10^5 \text{ m/s}$
2. There must be Homogenously concentrated fixed singly ionized (+ve) ions having concentration 'n' within a parallel homogenous electron beam of a circular cross section having concentration ' n_0 ' so that radius of the beam remains constant if the beam moves with constant velocity v along its axis.
- Find $\left(\frac{n}{n_0}\right)$?
- (A) $1 - \frac{v^2}{c^2}$ (B) $1 + \frac{v^2}{c^2}$ (C) $\frac{v^2}{c^2}$ (D) $\frac{c^2}{v^2}$
-

Space for Rough Work

3. Figure shows an isosceles right triangular thin sheet of uniform surface charge density σ . The self energy of the sheet is U_0 . What will be self energy of the sheet if it is folded along the dotted line.



- (A) $2U_0$ (B) $2\sqrt{2}U_0$ (C) $\sqrt{2}U_0$ (D) $\frac{1}{\sqrt{2}}U_0$
4. With an aim of determining the magnetic moment of a permanent magnet a simple experimental set up has been used in which a small cylindrical magnet of radius r_0 having magnetic moment M is placed at the center of a circular coil of radius r and having n turn of wires. The length of the magnet is perpendicular to the plane of the coil. Further the coil is connected to a ballistic galvanometer. The total resistance of the coil, connecting wires and the galvanometer is R . On sudden removal of the magnet from its position a charge q flows through the galvanometer. Find the magnetic moment of the magnet. ($r \gg r_0$)



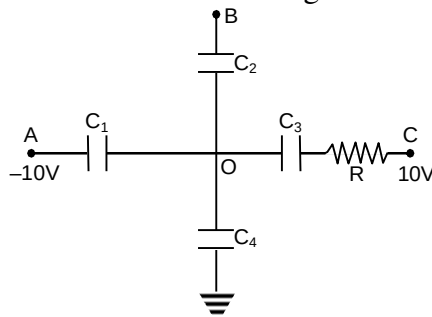
- (A) $\frac{qrR}{\mu_0 n}$ (B) $\frac{qr_0 R}{2\mu_0 n}$ (C) $\frac{2qrR}{\mu_0 n}$ (D) $\frac{2qr_0 R}{\mu_0 n}$

Space for Rough Work

SECTION- I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2 marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 marks;
 choosing **ONLY** (B) will get +1 marks;
 choosing **ONLY** (D) will get +1 marks;
 choosing no option (i.e. the question is unanswered) will get 0 marks, and
 choosing any other combination of options will get -1 mark.

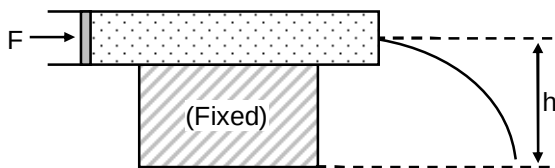
5. Figure shows part of a large network in which potentials of some of the points are shown. Each capacitor has capacitance $5\mu\text{F}$. Which of the following statements is/are true at steady state?



- (A) From the given information potential of point O can be determined but that of B cannot be determined
- (B) From the given information, potential of both points O and B cannot be determined
- (C) If charge on capacitor C_2 were also specified, potential of point O can be determine but that of B cannot be determined
- (D) If charge on capacitor C_2 were also specified, potential of both points O and B can be determined

Space for Rough Work

6. An ideal fluid of density ρ is filled in a horizontally fixed syringe fitted with piston of area A . An orifice is made at one end of the syringe. When piston is pushed into the syringe, the liquid comes out of the orifice and then following a parabolic path falls on the ground. Choose the correct statements.



(A) Liquid comes out of the orifice at speed $\sqrt{\frac{2(F + \rho ghA)}{\rho A}}$

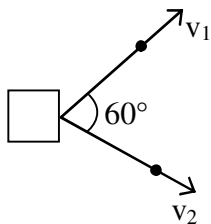
(B) Liquid comes out of the orifice at speed $\sqrt{\frac{2F}{\rho A}}$

(C) Liquid strikes the ground with speed $\sqrt{\frac{2(F + \rho ghA)}{\rho A}}$

(D) Liquid strikes the ground with speed $\sqrt{\frac{2F + \rho ghA}{\rho A}}$

Space for Rough Work

7. A block placed on ground is being pulled horizontally pulled by two persons with the help of inextensible strings. At a certain instant, velocities of the two persons are v_1 and v_2 as shown along the respective strings with strings making angle 60° with them



(Top view)

Choose the correct option

- (A) Speed of block at shown instant is $\frac{2}{\sqrt{3}} \sqrt{v_1^2 + v_2^2 - v_1 v_2}$
- (B) Speed of block at shown instant is $\frac{\sqrt{3}}{2} \sqrt{v_1^2 + v_2^2 - v_1 v_2}$
- (C) The instantaneous velocity of block will be along v_1 if $\frac{v_1}{v_2} = 2$
- (D) The instantaneous velocity of block will be along v_1 if $\frac{v_1}{v_2} = \frac{1}{2}$

Space for Rough Work

8. A glass capillary tube closed at upper end has internal and external radius as r and R respectively. The tube is held vertical with its lower end touching the surface of water. Assume that water perfectly wets the glass. Then, (Atmospheric pressure is P_0 , surface tension of water is T , density of water is ρ , mass of empty tube is M)

(A) For water in it to rise upto height h ($< L$), length (L) of such tube is $\left(h + \frac{2P_0rh}{2T - \rho gh} \right)$

(B) For water in it to rise upto height h ($< L$), length (L) of such tube is $\left(h + \frac{P_0rh}{2T - \rho gh} \right)$

(C) The vertical force needed to hold tube in this position will be

$$F = Mg + \pi P_0 \left[R^2 - \frac{Lr^2}{L - h} \right] + 2\pi(R + r)T$$

(D) The vertical force needed to hold tube in this position will be

$$F = Mg + \pi P_0 \left[R^2 - \frac{Lr^2}{L + h} \right] + 2\pi(R + r)T$$

9. Largest and smallest distance of a satellite from center of planet are given by $R_{\max}$ and $R_{\min}$ respectively. At these two positions speeds are given $v_{\min}$ and $v_{\max}$ respectively. The time period of revolution of the satellite is T . If 'a' and 'b' are semi major and semi minor axis then

(A) $a = \frac{R_{\max} + R_{\min}}{2}$

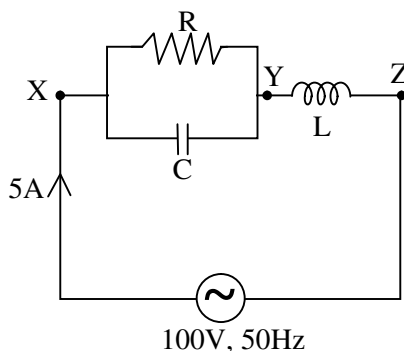
(B) $b = \sqrt{R_{\max} \cdot R_{\min}}$

(C) $a = \frac{T}{2\pi} \sqrt{v_{\max} v_{\min}}$

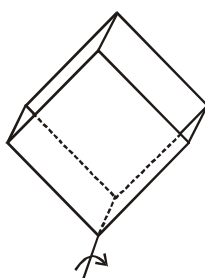
(D) $b = \frac{T}{2\pi} \sqrt{v_{\max} v_{\min}}$

Space for Rough Work

10. In an ac circuit shown below, voltages V_{XY} and V_{YZ} are 100 V each. If the main line current is 5A, then, [All given values are RMS values]



- (A) power consumed in circuit is $250\sqrt{3}$ watt.
 (B) The capacitive reactance of circuit will be 40Ω .
 (C) If $V_{\text{source}} = 100\sqrt{2} \sin(100\pi t)$, then current through resistor $I_R = \frac{5\sqrt{3}}{2} \sin\left(100\pi t - \frac{\pi}{3}\right)$
 (D) Current in capacitor C is lagging voltage V_{YZ} by 30° .
11. A solid cube of side length ℓ and mass M is pivoted on a horizontal axis such that it can rotate about one of its edges. Initially, it is held in the highest position and released. At the moment when the reaction of pivot towards centre of the cube becomes zero,



- (A) the horizontal component of reaction of pivot will be $\frac{3}{25} Mg$.
 (B) the horizontal component of reaction of pivot will be $\frac{4}{25} Mg$.
 (C) Angular acceleration of cube will be $\frac{6g}{5\sqrt{2} \ell}$
 (D) Angular velocity of cube will be $\sqrt{\frac{6g}{5 \ell}}$

Space for Rough Work

12. Consider water of Bulk Modulus β in the ocean having density ρ_0 , at surface at atmospheric pressure P_0 .

(A) Density ρ as a function of depth h is given as $\rho = \rho_0 \left(1 + \frac{\rho_0 g h}{\beta} \right)$

(B) Density ρ as a function of depth h is given as $\rho = \rho_0 \left(1 + \frac{\rho_0 g h}{2\beta} \right)$

(C) Pressure P as a function of depth is given as $P = P_0 + \rho_0 g h \left(1 + \frac{\rho_0 g h}{2\beta} \right)$

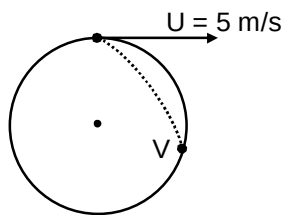
(D) Pressure P as a function of depth is given as $P = P_0 + \rho_0 g h \left(1 + \frac{\rho_0 g h}{\beta} \right)$

Space for Rough Work

SECTION- II : (Maximum Marks: 18)

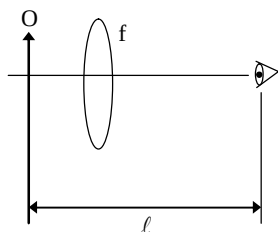
- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. When placed in a tank of liquid of density $\frac{3\rho}{2}$, the ball made of density ρ and 2ρ floats while remaining completely submerged. The time period of small angular oscillations of ball about its equilibrium position is given as $2\pi\sqrt{\frac{xR}{g}}$. Find x ?
2. When the gap is closed without placing any object in the screw gauge whose least count is 0.005 mm, the 5th division on its circular scale coincides with the reference line on main scale, and when a small sphere is placed reading on main scale advances by 4 divisions, whereas circular scale reading advances by 5 times to the corresponding reading when no object was placed. There are 200 divisions on the circular scale. The diameter of the sphere is $(100x) \mu\text{m}$. Find $\frac{x}{2}$.
3. A fixed sphere of radius $R = 3 \text{ m}$ is given. A small body is projected with speed $= 5 \text{ m/s}$ in horizontal direction from top point of inner surface of sphere. It hit inner surface of sphere as shown in diagram. The speed of body just before hitting inner surface of sphere is $V \text{ m/s}$. Fill the value V^2 in OMR. (Take $g = 10 \text{ m/s}^2$)

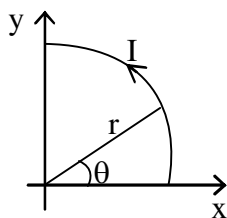


Space for Rough Work

4. The distance between object and eye is fixed equal ℓ . A convex lens of focal length f is to be placed in between object and eye. The distance between object and lens is less than focal length f . The maximum possible angular magnification is given as $\left(\frac{1}{1 + \frac{\ell}{kf}} \right)$. Find k



5. Two identical pendulums are suspended from the same point. They are deviated by same angle θ , they collide elastically at lowest point. After collision one of them stops and another just completes the vertical circle. The mass of ball (in kg) coming to rest if the mass of other ball is 3 kg.
6. A wire carrying current I is laid in shape of a curve which is represented in plane polar coordinate system as $r = 1 + \left(\frac{2}{\pi} \right) \theta$ for $\left(0 \leq \theta \leq \frac{\pi}{2} \right)$ here, θ is the angle measured with respect to positive x direction in anticlockwise sense and r is distance from origin. Magnetic field at the origin due to the wire is $\frac{\mu_0 I}{8} \ell \text{ n p}$. Find P



Space for Rough Work

PART-2: CHEMISTRY

SECTION-I(i): (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

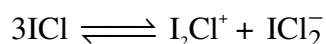
Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

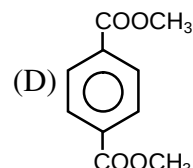
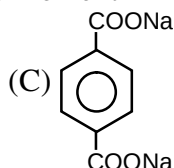
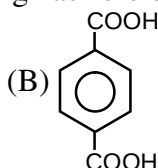
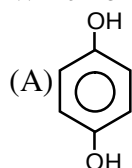
1. The acid-base reaction in non-aqueous solvents are typically described by means of the solvent system definition : acids are the compounds that increase the concentration of the solvonium (positive) ions, and bases are the compounds that result in the increase of the solvate (negative) ions, where solvonium and solvate are the ions found in the pure solvent due to self-dissociation of the solvent.

The auto-dissociation of ICl is given by the equation:



In ICl as solvent, AlCl_3 acts as

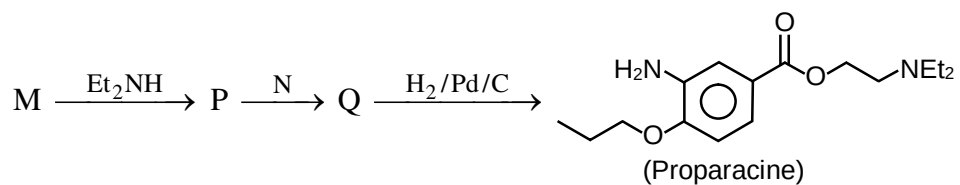
- (A) an acid (B) a base
(C) both acid as well as base (D) neither acid nor base
2. Which of the following has zero dipole moment-



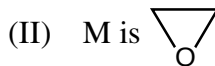
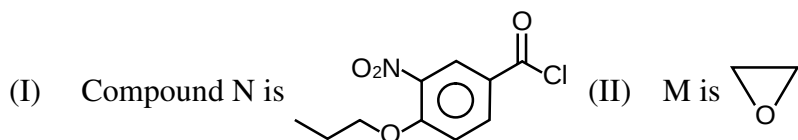
3. Which of the following has been named correctly?
- (A) $\text{H}_2\text{P}_2\text{O}_5^{2-} \Rightarrow$ dihydrogen pyrophosphite ion
(B) $\text{H}_3\text{P}_2\text{O}_5^- \Rightarrow$ trihydrogen pyrophosphate ion
(C) $\text{H}_2\text{PO}_3^- \Rightarrow$ dihydrogen phosphite ion
(D) $\text{H}_3\text{P}_2\text{O}_7^- \Rightarrow$ trihydrogenpyrophosphate ion

Space for Rough Work

4. Anaesthetic proparacine is made by this sequence of reactions:



Which statements are correct for following conversion?



(III) Mechanism for step $P \rightarrow Q$ is S_N2Ar

(IV) Compound M is $\text{CH}_3\text{-CH=O}$

(A) I, II & III

(B) II & III

(C) I & II

(D) I, II, III & IV

Space for Rough Work

SECTION-I(ii): (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2 marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 marks;
 choosing **ONLY** (B) will get +1 marks;
 choosing **ONLY** (D) will get +1 marks;
 choosing no option (i.e. the question is unanswered) will get 0 marks, and
 choosing any other combination of options will get -1 mark.

5. In which of the following cases the mentioned state will show degeneracy > 1 ?

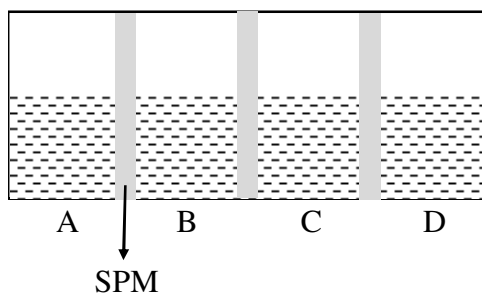
- (A) First excited state of He (B) First excited state of He^+
 (C) Third excited state of Li^+ (D) 2nd excited state of Li^{2+}

6. In which of the following $\Delta S_{\text{surrounding}}$ will drive the reaction in forward direction ?

- (A) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ (B) $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$
 (C) $\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{O}(\text{s})$ (D) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$

Space for Rough Work

7.

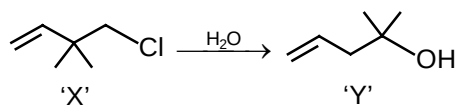


Four containers are connected using SPM(semipermeable membrane) having 100 ml 0.1M urea (A), 100 ml 0.1M NaCl (aq) (B), 100 ml 0.1M MgCl_2 (aq) (C) & 100 ml 0.1 M AlCl_3 (aq) (D). Assuming 100% ionisation of all electrolytes and no hydrolysis of ions, which of the following are correct?

- (A) Ratio of final volume of the four solutions after a very long time will be 1:2:3:4 respectively for A, B, C, D.
- (B) Final volume after a very long time will decrease in container A & B and will increase in C & D
- (C) Final volume of solution A will become 80 ml
- (D) Final volume of solution D will become 160 ml
8. Consider separation of galena (specific gravity 7.5) from siliceous gangue (specific gravity 2.7) using froth floatation process. Which of the following options is/are correct?
- (A) Silica particles rise with the froth as they are lighter than galena particles
- (B) Galena particles are floated to the froth with the help of sodium ethyl xanthate
- (C) Silica particles (SiO_2) are wetted by water molecules
- (D) Silica particles drown and are removed from the bottom of the floatation container
9. Among the following, select the groups in which all species will release energy by gaining an electron ?
- (A) Na(g) , P(g) , $\text{Cs}^+(\text{g})$ (B) $\text{Ca}^{2+}(\text{g})$, B(g) , Al(g)
- (C) He(g) , N(g) , $\text{S}^-(\text{g})$ (D) F(g) , Cl(g) , As(g)

Space for Rough Work

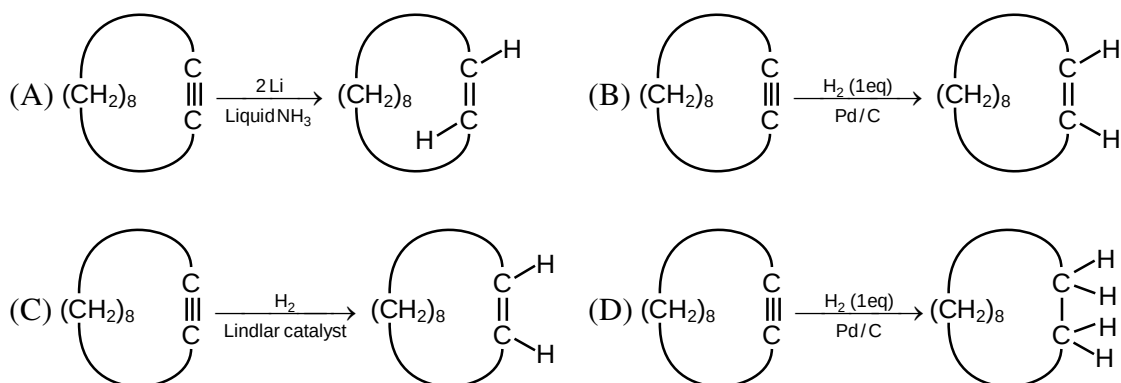
10. It has been observed that hydrolysis of 'X' occurs much faster than primary chlorides and it gives mainly 'Y'



Which of the following intermediate/s is/are involved in the above reaction?



11. Which of the following statement/s are correct for E1cB mechanism in alkyl halide?
- (A) E1cB rate law is identical to E2 rate law.
- (B) There is anti stereochemical preference in E1cB as base abstract proton anti to leaving group.
- (C) There is no stereochemical preference in E1cB because carbanion loses its stereochemical history through rapid inversion.
- (D) When the reaction is run in a deuterated medium the reversible deprotonation leads to accumulation of deuterated substrate.
12. In which of the following reactions correct major product is/are mentioned?



Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. How many moles of gases are evolved at room temperature by 4F charge in Kolbe's electrolysis of potassium propanoate (F = Faraday constant, current efficiency = 100% & consider only major products)
2. Air is taken in a 5L container fitted with a massless & frictionless piston at 27°C & 1 atm. External pressure is constant at 1 atm. Now a piece of U^{238} is also placed in the container. After a very long time volume of the container is found to be 15L. (Assume temperature constant & volume occupied by U^{238} is negligible) Final disintegration product of U^{238} is $^{206}_{82}\text{Pb}$ by α & β emission. Determine the energy needed in kJ (Rounded off to nearest hundred) to produce the same number of electron from H atom sample as produced in radioactive disintegration.
3. By aerobic oxidation, 80% of Pb^{2+} in red lead is converted to Pb^{4+} , if empirical formula of the new solid is Pb_xO_y . Determine $x + y$.
4. How many of the following orders are correct?
 - (i) $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$: boiling point
 - (ii) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$: melting point
 - (iii) $\text{SnH}_4 > \text{GeH}_4 > \text{CH}_4 > \text{SiH}_4$: melting point
 - (iv) $\text{SnH}_4 > \text{GeH}_4 > \text{SiH}_4 > \text{CH}_4$: boiling point
 - (v) $\text{NH}_3 > \text{SbH}_3 > \text{AsH}_3 > \text{PH}_3$: boiling point
 - (vi) $\text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$: melting point

Space for Rough Work

5. How many of the following relations is/are correct.
- $\text{Si-I} > \text{C-I}$ (Bond length)
 - $\text{Br}_2 > \text{F}_2$ (Bond dissociation energy)
 - $\text{Te} > \text{O}$ (Magnitude of $\Delta_{\text{eg}} H$)
 - $\text{N}_2 > \text{C}_2$ (Number of electrons in π Molecular orbitals)
 - $\text{He}^+ > \text{H}$ (Size)
 - N_2 Molecule $>$ N atom (Ionisation energy)
 - O_2 Molecule $>$ O atom (Ionisation energy)
 - $\text{O}_2^+ > \text{O}_2$ (Bond energy)
 - $\text{C}_2 > \text{O}_2$ (Bond distance)
 - $\text{Si-I} < \text{C-I}$ (Bond dissociation energy)
6. How many statements are correct for carbene?
- Singlet carbene has both electrophilic and nucleophilic properties, the balance between which is determined by the nature of group attached to carbon.
 - All electrons are paired in singlet carbene it can only take one orientation in a magnetic field (singlet).
 - Singlet carbene has characteristic of both a carbocation (empty p orbital) and carbanion (non bonding pair on carbon)
 - We might write abbreviated structure of a singlet carbene with + and – signs.

$$\begin{array}{c} \text{R} \\ \diagdown \\ \text{C}^\pm \\ \diagup \\ \text{R} \end{array}$$
 - Two unpaired electrons can adopt three different orientations in a magnetic field, a carbene of this structure is said to be in its triplet state.
 - Triplet carbene often reacts in ways that resemble free radicals.
 - For many carbenes such as $\text{:}\ddot{\text{C}}\text{H}_2$ the triplet carbene is of higher energy (Hence more unstable).
 - For carbene like :CCl_2 triplet is more stable.
 - $\text{CH}_2=\text{N}=\ddot{\text{N}}^+ \xrightarrow{\Delta} \ddot{\text{C}}\text{H}_2 + \text{N}_2$
 Diazo precursor has all its electron paired thermal decomposition leads to a singlet carbene.
 - $\text{CH}_2=\text{N}=\ddot{\text{N}}^+ \xrightarrow{h\nu} \ddot{\text{C}}\text{H}_2 + \text{N}_2$
 Photolysis can lead to either the singlet or triplet carbene depending on whether the photoexcited electronic state of the precursor is singlet or triplet.

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 12)**

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. 16 players $P_1, P_2, \dots, P_{16}$ play a knock out tournament. Assuming that the players are paired at random in each round. If all the players are of equal strength, then the probability that P_1 reaches final under the condition P_2 reaches semifinal but not in final is-
(A) $\frac{1}{15}$ (B) $\frac{2}{15}$ (C) $\frac{4}{15}$ (D) $\frac{8}{15}$
 2. If the curves $y = 2(x - a)^2$ and $y = e^{2x}$ touches each other, then 'a' is less than-
(A) -3 (B) -2 (C) -1 (D) 0
 3. If normal to the curve $y = x^2$ forms the shortest chord, then the square of slope of normal is equal to-
(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\sqrt{2}$ (D) 2
 4. Let $N = 2 \times 3 \times 5 \times 7 \times 11$ and $a \in \{1, 2, 3, \dots, N\}$, then the probability that 'a' is coprime with N is-
(A) $\frac{16}{77}$ (B) $\frac{32}{77}$ (C) $\frac{8}{77}$ (D) $\frac{64}{77}$
-

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2 marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 marks;
 choosing **ONLY** (B) will get +1 marks;
 choosing **ONLY** (D) will get +1 marks;
 choosing no option (i.e. the question is unanswered) will get 0 marks, and
 choosing any other combination of options will get -1 mark.

5. Let two curves $C_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (where $a > b$) and $C_2 : \frac{x^2}{c^2} - \frac{y^2}{d^2} = 1$ intersect at points P_1, P_2, P_3 and P_4 in Ist, IInd, IIIrd and IVth quadrant respectively. If P_1P_4 and P_3P_2 are latus rectum for both C_1 and C_2 , then-
- (A) $(P_1S_1)(P_1S_2)$ is equal to $(a^2 - c^2)$ where S_1, S_2 are foci of C_1
- (B) $e_1e_2 = 1$ (where e_1 and e_2 are eccentricities of C_1 and C_2 respectively)
- (C) curves $\frac{x^2}{a^2} + \frac{y^2}{c^2} = 1$ and $\frac{x^2}{b^2} - \frac{y^2}{d^2} = 1$ are confocal
- (D) C_1 and C_2 intersect orthogonally

Space for Rough Work

6. If locus of point $P(z)$ which satisfies $\arg(3+z+\bar{z}+|z|i) = \frac{\pi}{4}$ is a part of conic C , then -
- (A) Length of latus rectum of C is 6 (B) Length of latus rectum of C is $3\sqrt{2}$
 (C) eccentricity of C is 2 (D) eccentricity of C is $2\sqrt{2}$
7. If (α, β, γ) is a point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at distance $2\sqrt{7}$ from the point $(1, -1, 0)$, then the coefficient of $x^{\alpha+\beta+\gamma}$ in the expansion of $\left(2x^2 - \frac{1}{x}\right)^9$ is divisible by
- (A) 21 (B) 36 (C) 28 (D) 24
8. Let z and w be two complex numbers such that $|z| \leq 1$, $|w| \leq 1$ and $|z + i\bar{w}| = |z - iw| = 2$ then-
- (A) Number of complex number z satisfying above condition are 2
 (B) $\operatorname{Re}(z) = \operatorname{Im}(w)$
 (C) $\operatorname{Im}(z) = \operatorname{Re}(w)$
 (D) $|z| = |w| = 1$

Space for Rough Work

9. Let $f(x) = \ln(2x - x^2) + \sin \frac{\pi x}{2}$, then-

- (A) Graph of f is symmetric about the line $x = 1$
- (B) Graph of f is symmetric about the line $x = 2$
- (C) maximum value of f is 1
- (D) minimum value of f does not exist

10. If $I(k) = \int_0^{\infty} \frac{\ln x}{x^2 + kx + k^2} dx$ ($k > 0$) then-

- (A) $I(k) > \frac{\ln k}{3k}$
- (B) $I(k) < \frac{\ln k}{4k}$
- (C) $I(1) = 0$
- (D) $I(1) < \frac{\pi}{\sqrt{3}}$

Space for Rough Work

11. Given $\vec{A}(x) = x\hat{i} + x\hat{j} + 5\hat{k}$, $\vec{B}(x) = \hat{i} + x\hat{j}$, $\vec{C}(x) = \hat{i} + \frac{1}{5}\hat{j} + x\hat{k}$. Let α, β, γ denotes three values of x for which $\vec{A}(x), \vec{B}(x)$ and $\vec{C}(x)$ are coplanar, then-
- (A) $[\alpha] + [\beta] + [\gamma] = -1$
 (B) $[\alpha] + [\beta] + [\gamma] = 0$
 (C) Global maximum value of volume of parallelepiped formed by three vectors whose coterminal edges are $\vec{A}(x), \vec{B}(x)$ & $\vec{C}(x)$ is 4 for $x \in [-2, 3]$
 (D) $(1 + \alpha)(1 + \beta)(1 + \gamma) = -4$. (where $[\cdot]$ denotes greatest integer function)
12. Consider a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ for which $f'(0) = \ln 2$ and $f(x + y) = 2^x f(y) + 4^y f(x)$ $\forall x, y \in \mathbb{R}$. Which of the following is (are) correct ?
- (A) $f(4) = 240$
 (B) $f'(2) = 24 \ln 2$
 (C) the minimum value of $y = f(x)$ is $-1/4$
 (D) the number of solution of $f(x) = 2$ is 1

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places.
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-
1. A child has n coins of Re. 1. Each day he buys exactly one of the three objects either pencil for Rs.1 or pen for Rs.2 or book for Rs.2. If S_n denotes the number of ways of spending n rupees, then the value of $\left(\frac{S_{11} - S_{10}}{S_9} + S_7 \right)$ is
 2. Suppose a cubic polynomial $f(x) = x^3 + px^2 + qx + 72$ is divisible by both $x^2 + ax + b$ and $x^2 + bx + a$ (where a, b, p, q are real constants and $a \neq b$), then $f(-1)$ is equal to-
 3. If z_1, z_2, z_3 are complex numbers such that $|z_1| = 5, |z_2| = 3, |z_3| = 4$, then the maximum value of $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ is-

Space for Rough Work

4. Given that $f(x)$ is a function defined on $\mathbb{R}$, satisfying $f(1) = 1$ and for any $x \in \mathbb{R}$, $f(x + 5) \geq f(x) + 5$ and $f(x + 1) \leq f(x) + 1$. If $g(x) = f(x) + 5 - x$, then $g(2013)$ is equal to-
5. Tangent is drawn at the point (x_i, y_i) on the curve $y = f(x)$, which intersects the x-axis at $(x_{i+1}, 0)$. Now again a tangent is drawn at (x_{i+1}, y_{i+1}) on the curve which intersects the x-axis at $(x_{i+2}, 0)$ and the process is repeated 'n' times. If $x_1, x_2, x_3, \dots, x_n$ form a G.P. with common ratio 'r' and curve $y = f(x)$ is a hyperbola with eccentricity equal to 'e' then $r + e^2$ is equal to-
6. $\triangle ABC$ is isosceles with $AB = AC$ and $\angle CAB = 106^\circ$. Point M is an interior point such that $\angle MBA = 7^\circ$ and $\angle MAB = 23^\circ$, then $\sin(\angle MBC)$ is equal to

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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